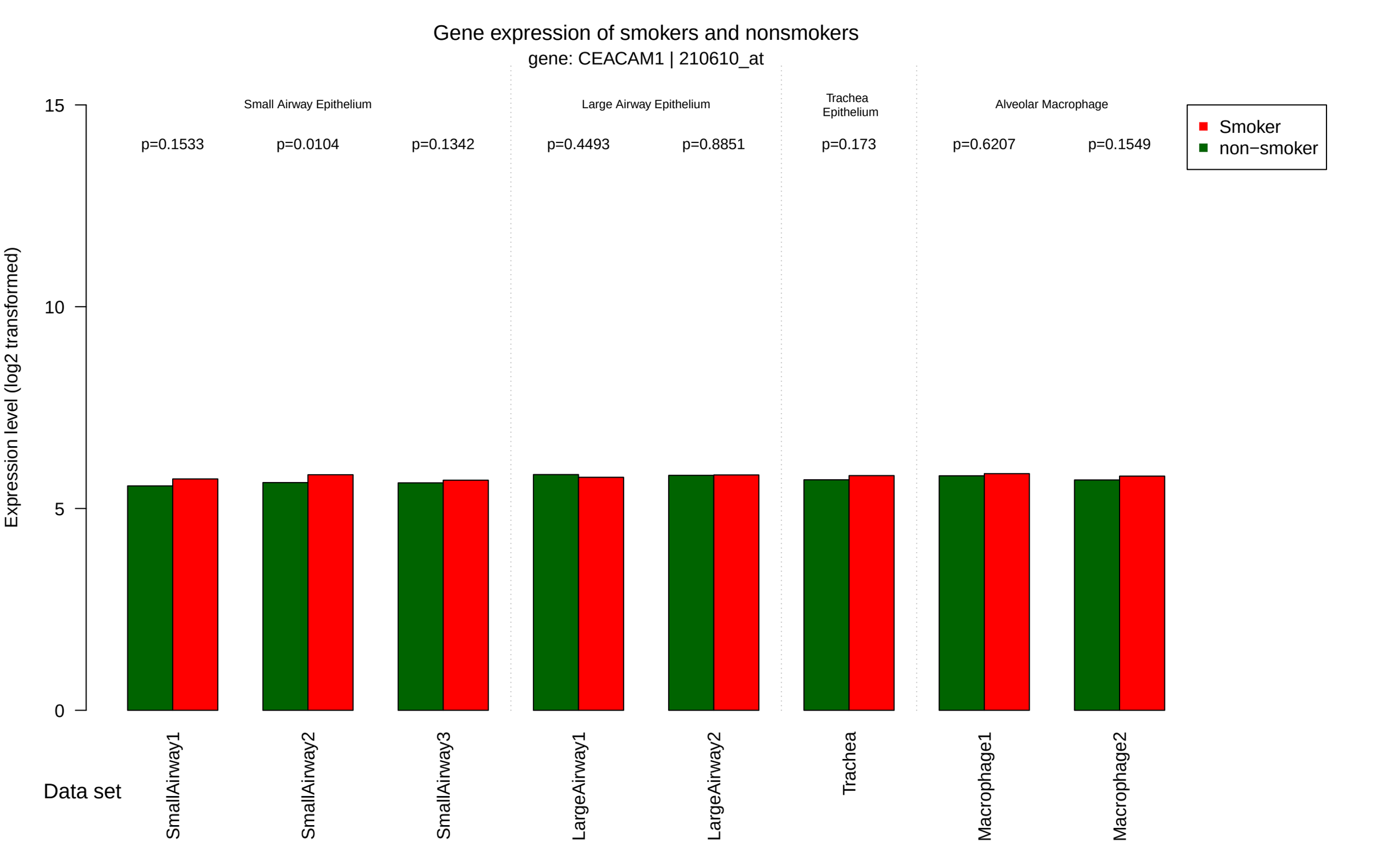
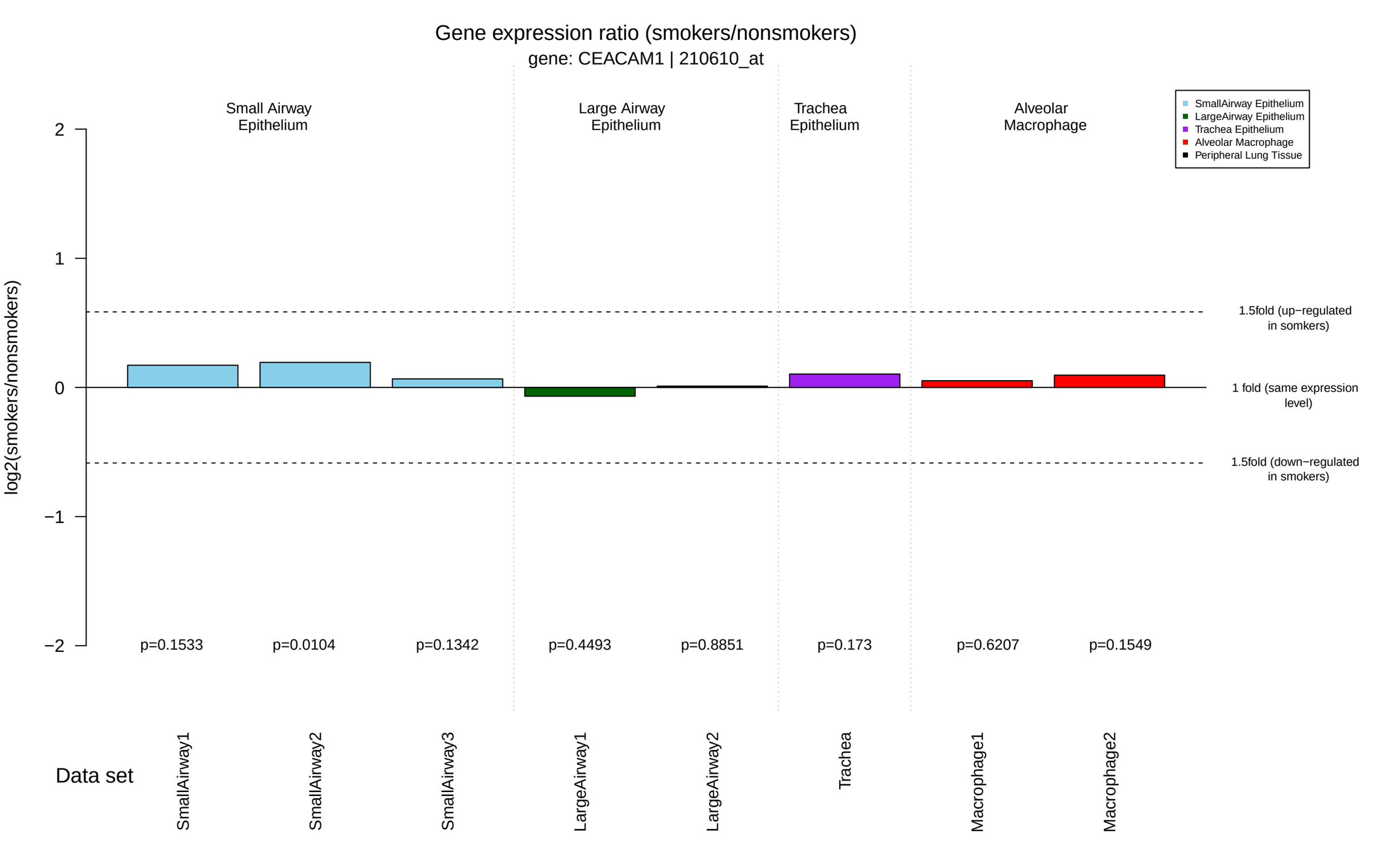
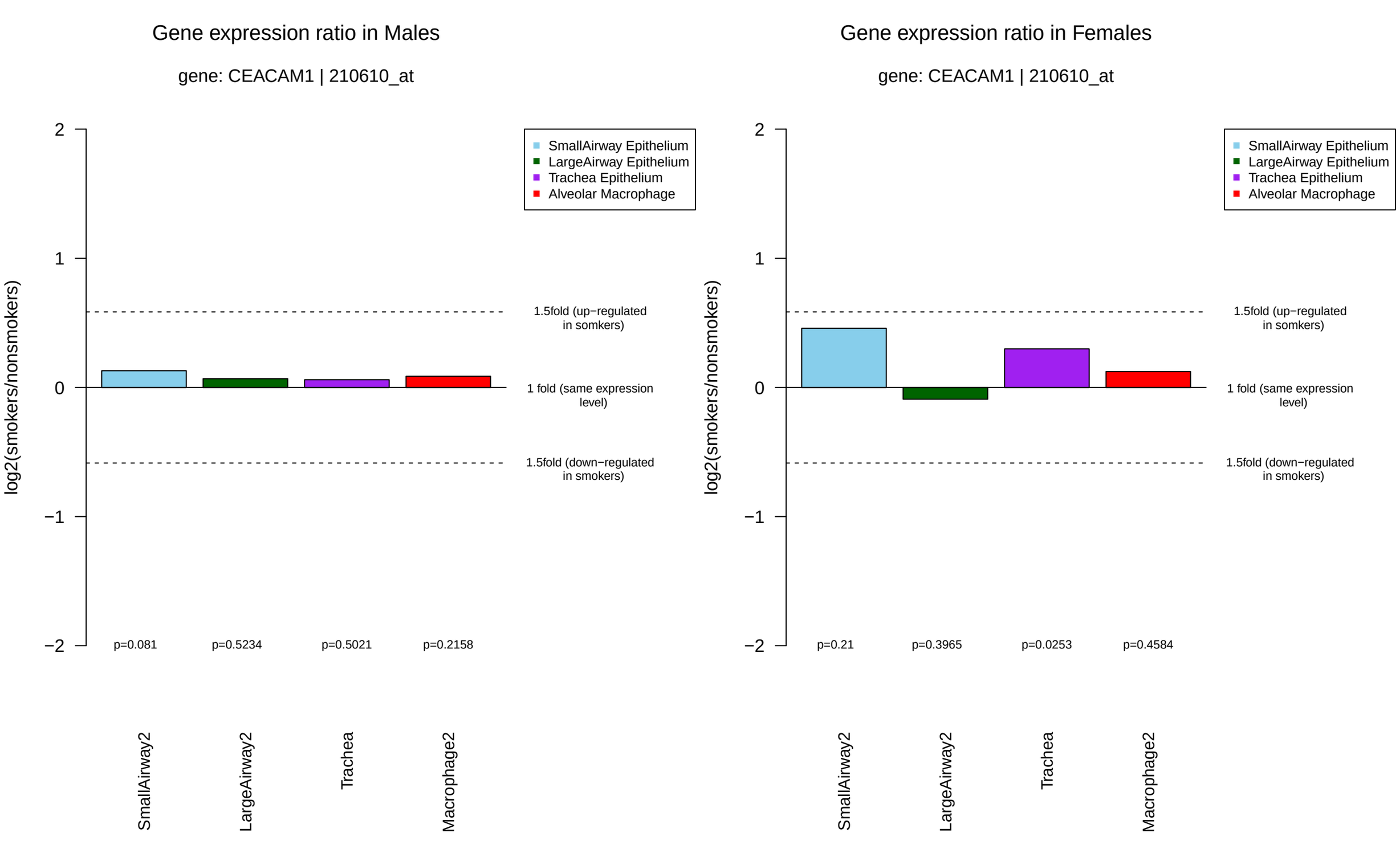
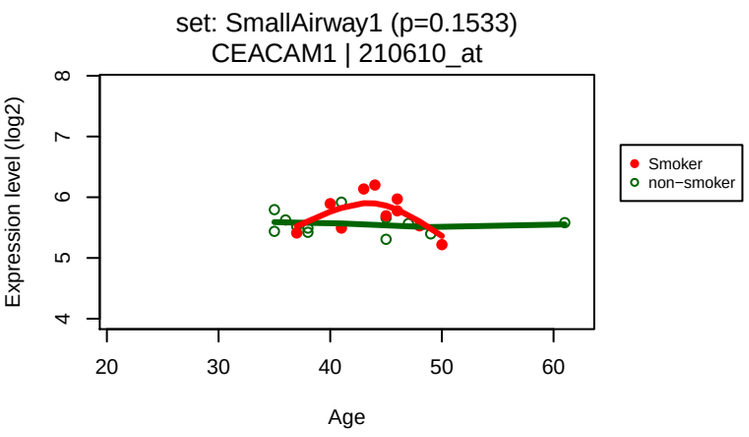
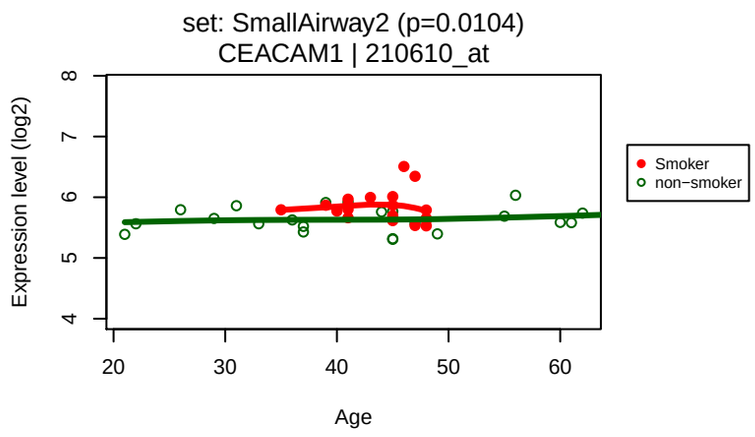
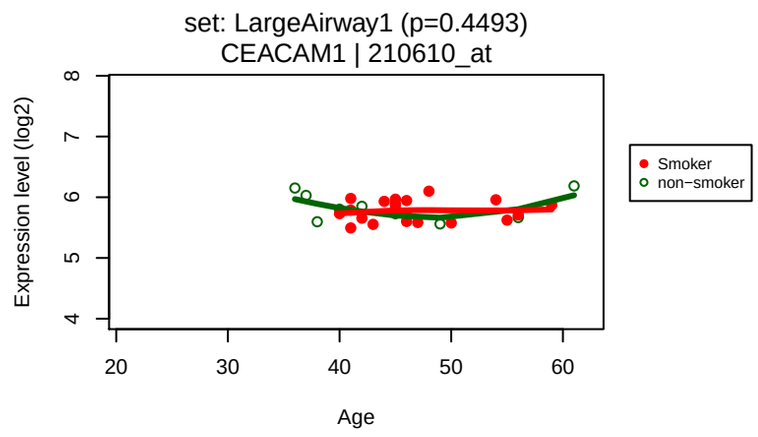

Gene expression of smokers and nonsmokers
gene: CEACAM1 | 210610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1533
p=0.0104
p=0.1342
p=0.4493
p=0.8851
p=0.173
p=0.6207
p=0.1549
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

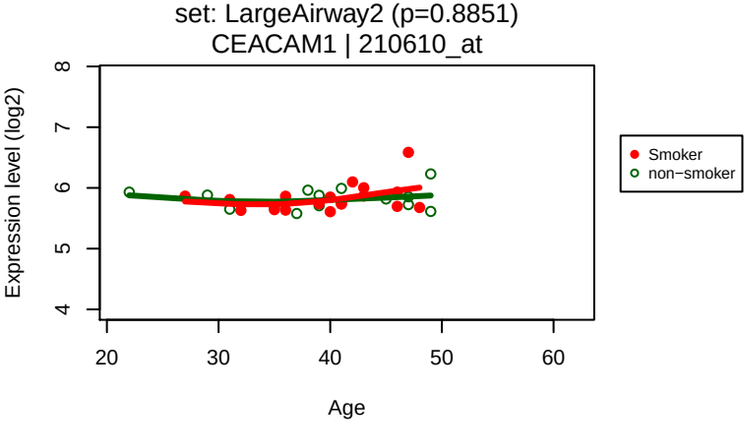

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM1 | 210610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1533
p=0.0104
p=0.1342
p=0.4493
p=0.8851
p=0.173
p=0.6207
p=0.1549
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

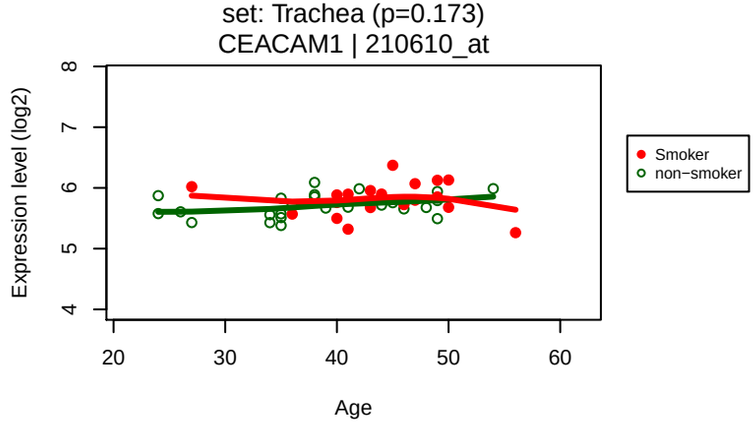

Gene expression ratio in Males
gene: CEACAM1 | 210610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.081
p=0.5234
p=0.5021
p=0.2158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM1 | 210610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.21
p=0.3965
p=0.0253
p=0.4584
SmallAirway2
LargeAirway2
Trachea
Macrophage2

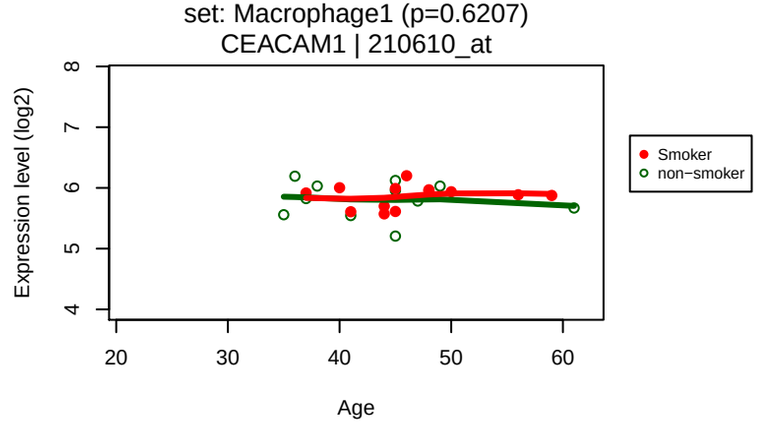

set: SmallAirway1 (p=0.1533)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker

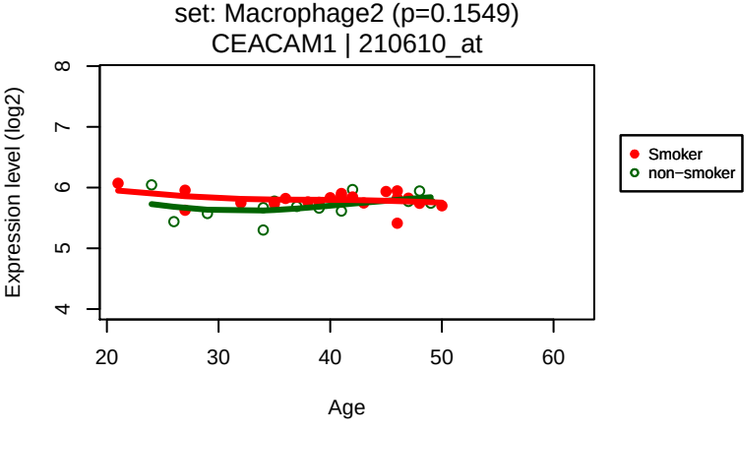

set: SmallAirway2 (p=0.0104)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker

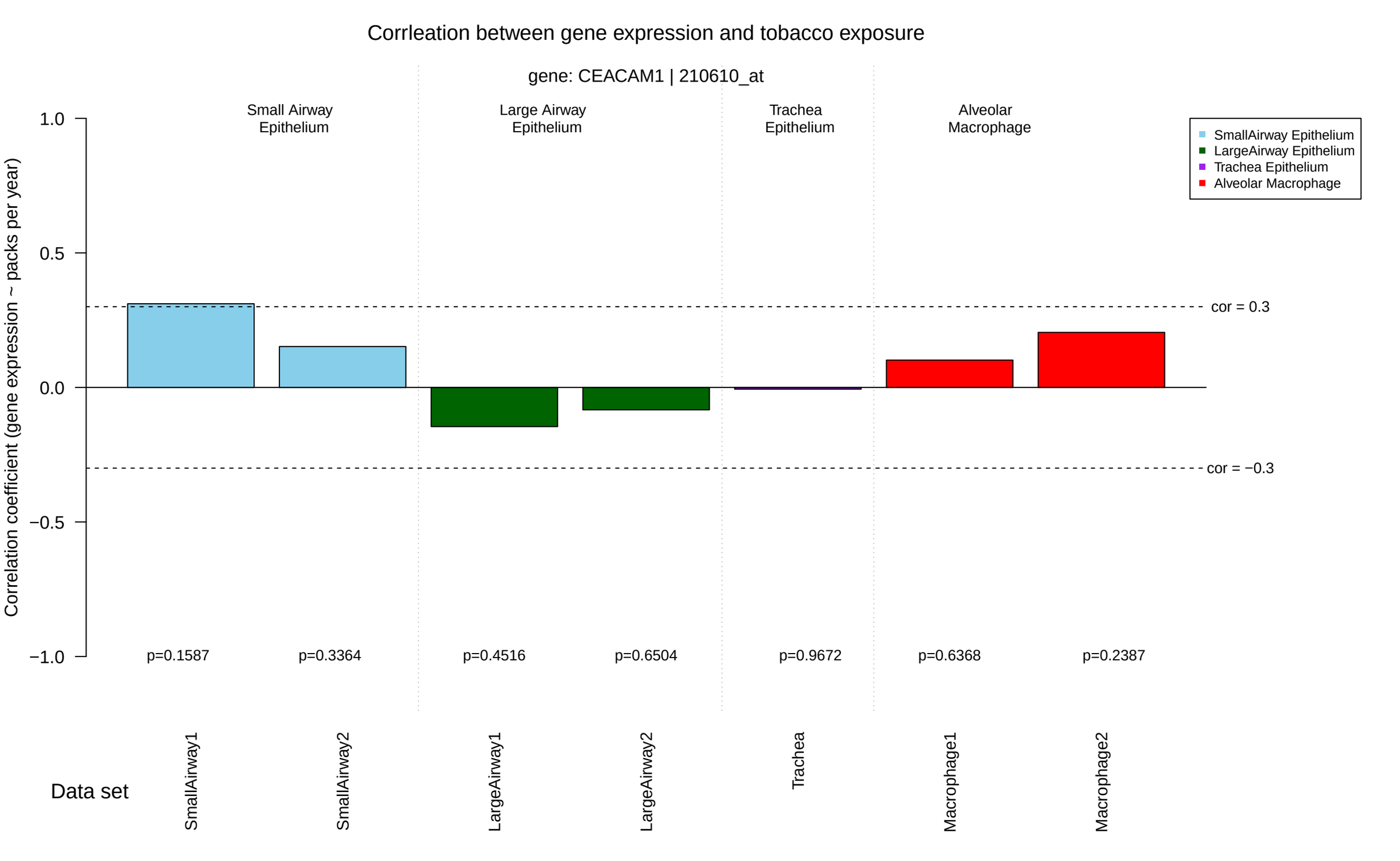

set: LargeAirway1 (p=0.4493)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8851)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.173)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6207)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1549)
CEACAM1 | 210610_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM1 | 210610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1587
p=0.3364
p=0.4516
p=0.6504
p=0.9672
p=0.6368
p=0.2387
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2